University of BRISTOL

COMS20018, Introduction to AI:

Philosophy of AI: LLMs, Rationality and Understanding

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Agenda

- We're going to look at some of the issues raised by Mike Woolridge in his lecture This is not the AI we were promised
- And some initiatives hoping to take AI further.

LLMs (and related systems) are game changers

- Similarly to Woolridge's talk we'll be looking at limitations of LLMs etc.
- But we're only interested in the limitations because they are so impressive.

> *There is only one thing in the world worse than being talked about, and that is not being talked about.*

The Picture of Dorian Gray, Oscar Wilde.

Well, what AI we're we promised?

- It is important to understand what a revolution modern generative AI is!
- AI used to be about building 'rational agents'.
- Contemporary AI (e.g. LLMs) is not like this.

Some LLM weirdness

- Woolridge gives many (amusing) examples or LLM irrationality.
- My friend Bryon Aragam also has some interesting examples from a NeurIPS 2024 paper. Let's have a look at it ...
- As Woolridge observes: GPT-5 still "hallucinates like crazy".
- Looks like eliminating hallucinations properly is a tough problem.

Sometimes the problems are more subtle

- Gemini on "What are Bayesian networks?" and "What is a Bayesian network?"

LLM reasoning

- One can, of course, show many examples of LLMs reasoning correctly.
- How do they do this?
- As Woolridge notes: natural language **encodes** logical reasoning and compositional structures. [my emphasis]
- But are LLMs rational minds?

Chain-of-thought reasoning

- Chain-of-thought reasoning is the state-of-the-art.
- Shifts effort to inference time.
- “Looks like algorithmic intelligence” - Woolridge.

Robots

- Robotic AI is the next threshold - Woolridge
- We still don't have robots that can do, say, household chores.
- So we have AI that replace some creative activities, but not the boring stuff - should be the other way round!
- Why is 'physical AI' harder than 'textual AI'?

AI in the UK

- "UKRI launches two AI research labs to stay ahead in global race"
- £60 million in funding
- UK government is looking for a niche for UK AI
- Wants to have 'AI sovereignty'

SOFAIR project

We're very excited that UCL will be leading the new SOFAIR Lab. While current AI systems are impressive, many still suffer from basic issues such as inaccurate responses to questions. These systems often use similar underlying architectures, so SOFAIR will bring together the broader sciences and fresh ideas to create a new generation of open-source models. This will reduce dependency on the small number of model providers, boosting UK sovereignty and its position as a global player in AI. (David Barber, UCL)

BOLD project

The UK cannot win the global AI race simply by trying to outspend the largest technology companies on data and compute. BOLD is about a different route: discovering fundamentally new ways to build AI that are more efficient, more open and better aligned with human needs. By focusing on new paradigms for learning, rather than only scaling existing methods, we aim to help secure the UK's sovereign capability in AI and ensure that academic research can shape the future of the field. (Jakob Foerster, University of Oxford)